

Question 1

NOT COMPLETE

Case Study Screen 1 of 6

The nurse in a pediatric primary care clinic is assessing Ethan, a 4-year-old boy brought in by his mother for frequent falls.

Nurses' NotesVital Signs & Growth

0915:

Ethan, a 4-year-old boy, is brought to the pediatric primary care clinic by his mother, Sarah (29 years old), because his preschool teacher reports that he 'falls more than the other children and cannot keep up on the playground.' Sarah states he walked independently at 18 months, has never been able to run well, and in the past 6 months has begun walking on his toes and needs to hold the railing with both hands to climb stairs. She adds that he has a mild expressive speech delay and is followed by a speech therapist.

0930:

Gait is wide-based and waddling with an exaggerated lumbar lordosis. When asked to stand up from sitting on the floor, Ethan rolls to a prone position, pushes up onto his hands and feet, and 'walks' his hands up his thighs to reach standing; timed rise from the floor is 7 seconds. Both calves are enlarged and feel firm and rubbery on palpation. Unable to hop on one foot. Heel cords are tight bilaterally with ankle dorsiflexion limited to neutral. Fine motor skills and hand grip are age-appropriate. No muscle tenderness, rash, or joint swelling.

0940:

Family history: Sarah's brother died at age 20 of 'heart and breathing failure from a muscle disease' that began when he was a young child. Ethan has a healthy 2-year-old sister, Lily. Sarah and her partner are planning another pregnancy within the next year. Immunizations are up to date for age. Primary care provider notified of findings.

The following 6 questions refer to this clinical scenario: NURS 1017 Unit 3 Case Study 4.

- The nurse reviews Ethan's history and physical assessment. Which findings should the nurse recognize as cues of a **progressive inherited neuromuscular disorder**? **Select all that apply.**
- ☐ 1. Uses his hands to push up on his thighs to rise from the floor, taking 7 seconds
 - ☐ 2. Heart rate 104 bpm and respiratory rate 24 breaths/min
 - ☐ 3. Enlarged, firm, rubbery calf muscles bilaterally
 - ☐ 4. Maternal uncle died at age 20 of heart and breathing failure from a childhood-onset muscle disease
 - ☐ 5. Age-appropriate hand grip and fine motor skills
 - ☐ 6. Waddling gait with exaggerated lumbar lordosis and difficulty climbing stairs
 - ☐ 7. Height at the 40th percentile and weight at the 50th percentile
 - ☐ 8. Walked independently at 18 months, with frequent falls and toe-walking

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CORRECT

Score: 5 / 5 points

Rationale:

Correct cues:

Gowers sign (climbing up the thighs with the hands to stand) reflects weakness of the proximal hip and thigh muscles, the first muscles affected in Duchenne muscular dystrophy (DMD).

Question 2

NOT COMPLETE

Case Study Screen 2 of 6

Ethan's laboratory and genetic test results are now available.

Nurses' Notes

Laboratory Results

Genetic Testing Report

1400:

Day 10: Ethan and Sarah return to the clinic to review laboratory results with the provider. The provider explains that a markedly elevated creatine kinase with a normal GGT suggests that the enzymes are leaking from damaged skeletal muscle rather than from the liver, and that genetic testing has confirmed a diagnosis of Duchenne muscular dystrophy. Sarah is tearful and asks, 'Did I cause this? My brother had the same thing.' Referral placed to the neuromuscular clinic. Sarah consents to carrier testing.

➤ The nurse compares Duchenne muscular dystrophy with Becker muscular dystrophy to prepare family teaching. For each feature, click to specify whether it is characteristic of **Duchenne muscular dystrophy**, **Becker muscular dystrophy**, or **Both**.

Feature	Duchenne	Becker	Both
Out-of-frame DMD gene deletion resulting in absent dystrophin protein	☐	☐	☐
In-frame DMD gene deletion producing a shortened, partially functional dystrophin protein	☐	☐	☐
X-linked recessive inheritance affecting mostly males	☐	☐	☐
Onset of weakness in early childhood with loss of independent walking by about age 12 without treatment	☐	☐	☐
Milder course, with many people still walking into their late teens or adulthood	☐	☐	☐
Elevated serum creatine kinase and risk of cardiomyopathy	☐	☐	☐

Note: Each row must have only 1 response option selected.

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CORRECT

Score: 6 / 6 points

Rationale:

Both disorders are **X-linked recessive** and are caused by mutations in the same gene, DMD, which codes for dystrophin. Dystrophin anchors the inside of the muscle fiber to its outer membrane and protects the membrane during contraction.

Duchenne: an out-of-frame deletion (such as Ethan's exons 45–50) shifts the genetic reading frame, so almost no functional dystrophin is made. Muscle membranes tear with everyday use, and fibers die and are replaced by fat and fibrous tissue. Weakness starts in early childhood, and without treatment children usually stop walking by about age 12.

Becker: an in-frame deletion keeps the reading frame intact, so a shorter but partly working

Question 3

NOT COMPLETE

Case Study Screen 3 of 6

Three weeks after diagnosis, Ethan's parents meet with the nurse and genetic counselor to review carrier test results.

Nurses' Notes

Family Genetics

Laboratory Results

1000:

Week 3: Genetic counseling visit with Sarah and her partner, Daniel. Sarah's carrier testing has resulted (see Family Genetics tab). Couple asks what the results mean for Lily and for a future pregnancy. Sarah states, 'Everyone keeps asking about Ethan's liver. Is his liver failing too?'

➤ The nurse reinforces the genetic counselor's teaching. Complete the following sentences by choosing from the lists of options.

Because Sarah is a heterozygous carrier, each of her sons has a

Select...▼

 chance of being affected by DMD, and each daughter has a

Select...▼

 chance of being a carrier. Ethan's elevated AST and ALT with a normal GGT and bilirubin most likely reflect

Select...▼

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➤ The nurse reinforces the genetic counselor's teaching. Complete the following sentences by choosing from the lists of options.

Because Sarah is a heterozygous carrier, each of her sons has a

50%

 chance of being affected by DMD, and each daughter has a

50%

 chance of being a carrier. Ethan's elevated AST and ALT with a normal GGT and bilirubin most likely reflect

enzyme release from damaged skeletal muscle

.

Submit

CORRECT

Score: 3 / 3 points

Rationale:

X-linked recessive inheritance:

a carrier mother has one normal X and one X with the DMD mutation, and passes one of them to each child at random.

Sons

get their only X from their mother (and a Y from their father), so each son has a **50% chance** of inheriting the mutated X and being affected.

Daughters

get one X from each parent. Each daughter has a **50% chance** of receiving the mutated X and being a carrier; because she also has her father's normal X, she is usually unaffected. Carrier women do have a risk of cardiomyopathy and should have periodic heart screening.

Liver enzymes:

AST and ALT are found in skeletal muscle as well as the liver. When they rise alongside a very high CK while liver-specific markers (GGT, bilirubin) are normal, the source is damaged muscle, not the liver. Recognizing this prevents unnecessary liver work-ups such as a liver biopsy.

Question 4

NOT COMPLETE

Case Study Screen 4 of 6

Two months after diagnosis, the nurse is coordinating Ethan's multidisciplinary plan of care at the neuromuscular clinic.

Nurses' Notes

Care Plan Orders

1100: Month 2: Neuromuscular clinic visit. Ethan (now 4 y 3 mo, 17.6 kg) remains ambulatory; timed rise from floor 7.4 seconds. The neurologist plans to begin daily oral glucocorticoid therapy to slow the loss of muscle strength and discusses the multidisciplinary care plan. Chart review: Ethan received 1 dose of varicella vaccine at 12 months; varicella IgG titer pending.

➤ The nurse is planning care and teaching for Ethan and his family. Which interventions should the nurse include in the plan of care? **Select all that apply.**

- ☐ 1. Confirm varicella immunity and complete any outstanding immunizations before starting prednisone
- ☐ 2. Limit Ethan to bed and wheelchair rest to conserve remaining muscle strength
- ☐ 3. Teach the family a daily home stretching program for the heel cords and hips, and use night-time ankle-foot orthoses
- ☐ 4. Encourage swimming and tricycle riding, and avoid trampolines and repeated downhill running
- ☐ 5. Recommend a high-calorie diet to build muscle mass
- ☐ 6. Place an alert in Ethan's chart and provide a wallet card stating that succinylcholine and inhaled volatile anesthetics must be avoided
- ☐ 7. Recommend that Sarah, as a confirmed carrier, have periodic cardiac screening
- ☐ 8. Monitor serum creatine kinase every week to measure how fast the disease is progressing

Submit

Question 4

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Case Study Screen 4 of 6

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CORRECT

Score: 5 / 5 points

Rationale:

Immunizations first: daily glucocorticoids weaken the immune response. Varicella immunity should be confirmed and vaccines completed before steroid therapy begins, and annual inactivated influenza vaccination continues afterward.

Question 5

NOT COMPLETE

Case Study Screen 5 of 6

One year later, Ethan is brought to the emergency department after a fall and needs a fracture reduction under procedural sedation.

Nurses' Notes

Vital Signs

1615:

One year later: Ethan, now 5 years old, is brought to the emergency department after tripping at the park and falling onto an outstretched right hand. Right distal forearm is swollen and angulated; X-ray shows a displaced distal radius fracture. Radial pulse 2+, capillary refill < 2 seconds, fingers warm and able to wiggle. Pain 6/10 (FACES). Ethan has taken prednisone 13 mg daily for the past 10 months; last dose at 0730 today. Orthopedics plans closed reduction under procedural sedation. Ethan last ate at 1230.

➤ The emergency department nurse is preparing Ethan for closed reduction of his forearm fracture. For each nursing action, click to specify whether the action is **Indicated**, **Non-Essential**, or **Contraindicated**.

Nursing Action	Indicated	Non-Essential	Contraindicated
Tell the provider that Ethan has taken prednisone daily for 10 months, so a stress dose of steroids can be considered	☐	☐	☐
Hold Ethan's prednisone for the next several days while he is recovering	☐	☐	☐
Prepare succinylcholine as the paralytic agent in case airway management is needed	☐	☐	☐
Assess neurovascular status of the right hand before and after the reduction	☐	☐	☐
Obtain a serum creatine kinase level before the procedure	☐	☐	☐
Keep Ethan on strict bed rest for 2 weeks after the reduction	☐	☐	☐

Note: Each row must have only 1 response option selected.

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Question 5

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CORRECT

Score: 6 / 6 points

Rationale:

Indicated – stress-dose steroids: long-term glucocorticoid use suppresses the body's own adrenal response. Procedures and injuries raise cortisol needs, so the provider must know about Ethan's steroid use in order to consider stress dosing and prevent adrenal crisis (low blood pressure, shock).

Contraindicated – holding prednisone: stopping chronic steroids abruptly can cause adrenal insufficiency. The daily dose must continue unless the prescriber changes it.

Contraindicated – succinylcholine: in DMD it can cause rhabdomyolysis, severe hyperkalemia and

Question 6

NOT COMPLETE

Case Study Screen 6 of 6

Ethan returns to the neuromuscular clinic 18 months after starting prednisone for a routine evaluation.

Nurses' Notes

Follow-Up Data

0930:

18 months after starting prednisone: Routine neuromuscular clinic follow-up. Ethan (now 5 y 10 mo) is walking independently, and Sarah reports that falls have decreased from several per day to about 1 per week since starting prednisone. He attends kindergarten with an individualized education program (IEP). Sarah is concerned that he 'is always hungry,' has 'grown out of his clothes sideways,' and has more frequent temper outbursts.

➤ The nurse evaluates Ethan's response to the plan of care. Complete the following sentences by choosing from the lists of options.

Ethan's stable timed rise from the floor and 10-meter walk/run time indicate that prednisone therapy is

Select...

His weight gain, slowed growth in

height, and vitamin D level of 16 ng/mL indicate a need to

Select...

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When Sarah asks whether the prednisone can simply be stopped because of the weight gain, the nurse explains that it

Select...

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➤ The nurse evaluates Ethan's response to the plan of care. Complete the following sentences by choosing from the lists of options.

Ethan's stable timed rise from the floor and 10-meter walk/run time indicate that prednisone therapy is

achieving its goal of slowing the loss of muscle strength . His weight gain, slowed growth

in height, and vitamin D level of 16 ng/mL indicate a need to

request a dietitian referral and discuss vitamin D and calcium supplementation with the provider .

When Sarah asks whether the prednisone can simply be stopped because of the weight gain, the nurse explains that it

must never be stopped suddenly; any change must be a gradual taper directed by the provider .

Submit

CORRECT

Score: 3 / 3 points

Rationale:

Expected outcome: in DMD, steroids aim to slow the decline, not to cure. For a 5–6-year-old with Duchenne, holding his motor function steady (rise time 7.4 → 7.1 s) and falling less shows the treatment is working. It does not mean the treatment can stop; the underlying dystrophin deficiency remains.

Adverse effects needing a plan change: steroid-related weight gain (BMI now at the 98th percentile), slowed growth in height, and vitamin D deficiency. Steroid treatment and reduced mobility both weaken bones and raise fracture risk, as Ethan's broken arm showed. The nurse should arrange nutrition counseling and discuss vitamin D and calcium supplementation with the provider. Behavior changes and increased appetite are also common steroid effects that the family should know about.

Safety teaching: chronic glucocorticoids suppress the adrenal glands. Stopping them suddenly can cause a life-threatening adrenal crisis, so any change in dose must be a gradual taper ordered by the prescriber.